

You must show your work to receive credit. Unless stated otherwise, all problems are worth 5 points.

1. [2pts each] Find the following:

a. $x^5 - 10000$ as $x \rightarrow \infty$

b. $\lim_{x \rightarrow \infty} \frac{1}{x}$

c. $\lim_{x \rightarrow -3^+} \frac{2}{x+3}$

d. The domain of $\frac{3x^{13} - 12x + \sqrt{5}}{(x+3)(x-1)}$

For problems 2 through 7, find all real solutions to the following equations. Write all answer as exact values

2. $\frac{3x}{2} - 1 = 2 - \frac{7x}{4}$

5. $\sqrt{x+3} = x+1$

3. $\frac{1}{2}|x-9| - 7 = 9$

6. $2e^{x-1} = 10$

4. $x^2 - 11 = 0$

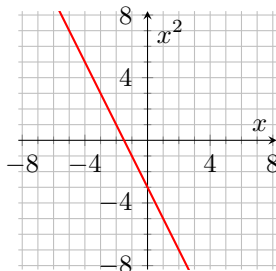
7. $\log_3(4) = 3 - \log_3(x)$

Solve the word problems. Approximate your answer to three decimal places.

8. Jeff knows that Holly paid \$40,230, including sales tax for a new Buick La-Sabre. If the sales tax is 10%, then what is the cost of the car *before* the tax?

9. How long will it take \$100 to grow to \$350 if it is invested at 8% compounded continuously?

10. Write an equation for the line shown in the graph.



11. [2 pts each] Perform the indicated operations and write your answers in the form $a + bi$.

a. $(2 - i)(4 + 8i)$

b. i^4

c. i^{25}

12. Let $f(x) = x + 1$ and $g(x) = x^2 - x$. Find and simplify the following expressions.

a. [2 pts] $g(-2)$

b. $(g \circ f)(x)$

13. [2pts each] Use the graph of the quadratic function, to answer the following:

a. What is the vertex?

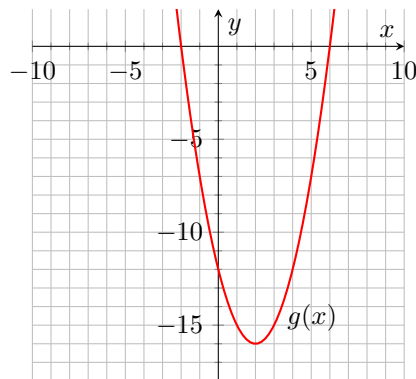
b. What is the range?

c. If any exist, find the x -int(s).

d. Find the y -int.

e. What are the zeros of $f(x)$?

f. Find the simplest polynomial with real coefficients that has the roots from above (you do not have to expand).



14. Consider the polynomials $P(x) = 2x^2 - 3x + 6$ and $(x - 5)$.

a. Use division (synthetic or traditional) to find the quotient and remainder when $P(x)$ is divided by $x - 5$.

b. [2 pts] Using your answer above, find $P(5)$.

c. [2 pts] Is $x = 5$ a root/zero of $P(x)$?

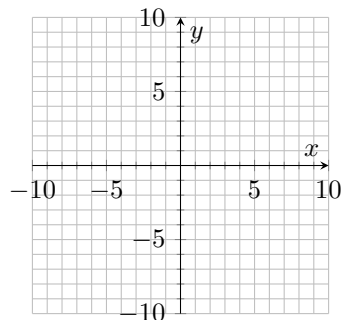
15. [2 pts each] Use $f(x) = \frac{2x + 1}{x + 3}$

a. Find the domain of $f(x)$.

b. Find any vertical asymptotes.

c. Find the horizontal asymptote.

d. Sketch a graph of the function



16. Use $\frac{f(x+h) - f(x)}{h}$ for $h \neq 0$, to complete and simplify the following problem: Find the difference quotient of $f(x) = 2x + 1$.

17. [3 pts each] Evaluate the following:

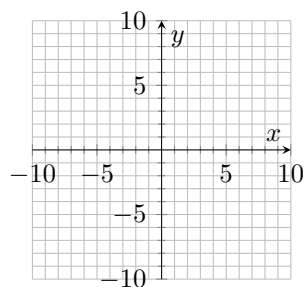
a. $\log_6(6^2)$

b. $\frac{1}{3}e^{\ln(3)}$

18. Let $g(x) = \log_2(x - 2)$.

- a. Graph $g(x)$.

- b. [2 pts] Find the domain and range of $g(x)$.



19. Use the properties of logarithms to expand the expression as much as possible.

$$\log_{11}\left(\frac{11}{y^2}\right)$$

20. Use the properties of logarithms to rewrite the expression as a single logarithm.

$$\ln(4) + 2\ln(3)$$

21. Find and sketch the graph of the equation of the parabola with focus $(-1, 6)$ and directrix $y = 2$.

